

BURHANUDDIN

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EDUCATION

Master of Science in Cybersecurity Sep 2026 – May 2028 (Expected)
New York University (NYU), Tandon School of Engineering, New York, NY

Bachelor of Technology, Computer Science and Engineering Aug 2020 – May 2024
Rajasthan Technical University (Techno India NJR Institute), India
GPA: 9.4 / 10 (First Division with Distinction)

TECHNICAL SKILLS

- **Security & Networking:** Socket Programming, TCP/IP, Cryptography, TCPdump, Burp Suite, Splunk
- **Programming & Scripting:** Python, C++, SQL, Bash
- **Systems & Architecture:** Linux (RHEL, Debian), Windows API, Process Management, OS File Descriptors
- **Cloud & Applied AI:** Azure AI Services, RAG Workflows, Prompt Engineering, Cloud Security Frameworks

EXPERIENCE & LEADERSHIP

Section Leader (Instructor) April 2026 – June 2026
Stanford University (Code in Place) | Remote | [Credential](#)

- Instructed a global cohort of 15 students in Python fundamentals, control flow, and data structures over a 6-week program, achieving an 85% early assignment completion rate.
- Led live debugging sessions, dry-running execution flows to teach practical troubleshooting and logic building.

PROJECTS & RESEARCH

TargetPortScanner | *Python, Socket API, I/O Multiplexing* | [GitHub](#) June 2026

- Engineered an event-driven TCP port scanner using the `selectors` module for non-blocking I/O multiplexing, cutting a full 65,535-port sweep from ~45 days (blocking 1-minute timeouts) to seconds, a 240,000x throughput gain.
- Implemented a min-heap priority queue to aggressively manage socket timeouts and cull dead connections, preventing memory leaks from stealth firewall packet drops.
- Designed a batch-processing architecture to bypass Windows OS file descriptor limits (`select()` constraints), ensuring stable execution during high-volume network polling.

WORM: Worm Behavior Simulator | *Python, Windows API, psutil* | [GitHub](#) December 2025

- Engineered a sandboxed Python simulator to study malware self-replication and system persistence strategies.
- Implemented stealth execution using Windows API (`ctypes`) to automate background replication and evade standard graphical interface detection.
- Developed a custom Endpoint Detection and Response (EDR) utility using `psutil` to systematically hunt rogue processes via command-line signatures, implementing dynamic exception handling for OS-level access denials.

PROFESSIONAL CERTIFICATIONS

Red Hat Certified System Administrator (RHCSA) | [Credential](#) April 2025
Red Hat – Linux System Administration, Kernel Management, Security

Microsoft Certified: Azure AI Engineer Associate (AI-102) | [Credential](#) March 2025
Microsoft – Azure AI Services, Microsoft Foundry, RAG, Custom Models

Google Cybersecurity Professional Certificate | [Credential](#) November 2024
Coursera – Network Analysis, Python for Security, Frameworks